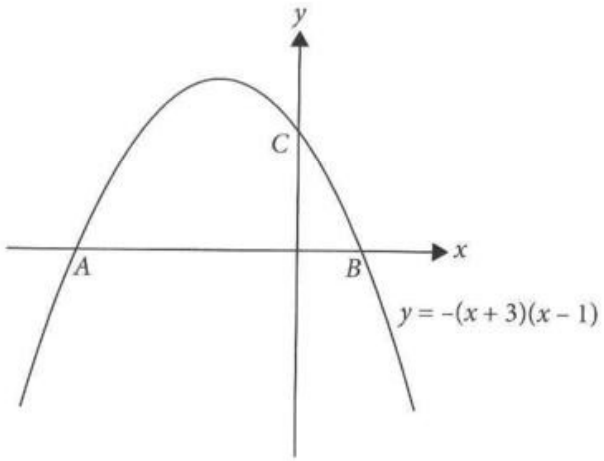
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Worksheet 5B

Quadratic functions and graphs

1. The diagram shows the curve $y = -(x + 3)(x - 1)$.
The curve cuts the x-axis at the points A and B, and the y-axis at the point C.

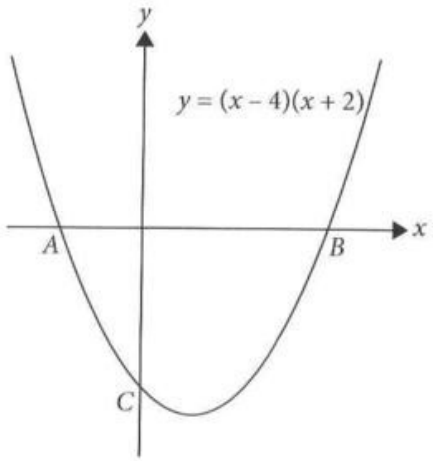


(i) Find the coordinates of A, B and C.

(ii) The equation of the line of symmetry is $x = -1$.
Determine the coordinates of the maximum point of the curve.

---

2. In the diagram, the curve $y = (x - 4)(x + 2)$ cuts the x-axis at the points A and B, and the y-axis at the point C.



(i) Find the coordinates of A, B and C.

(ii) The point $P(-3, k)$ lies on the curve.
Find the value of k.

(iii) Explain why the point $Q(3.5, -2.5)$ does not lie on the curve.

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3. The variables x and y are connected by the equation $y = x^2 - 3x - 2$.
Some corresponding values of x and y are given in the table below.

x	−2	−1	0	1	2	3	4
y	p	2	−2	−4	−4	−2	2

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-2 \leqslant x \leqslant 4$.

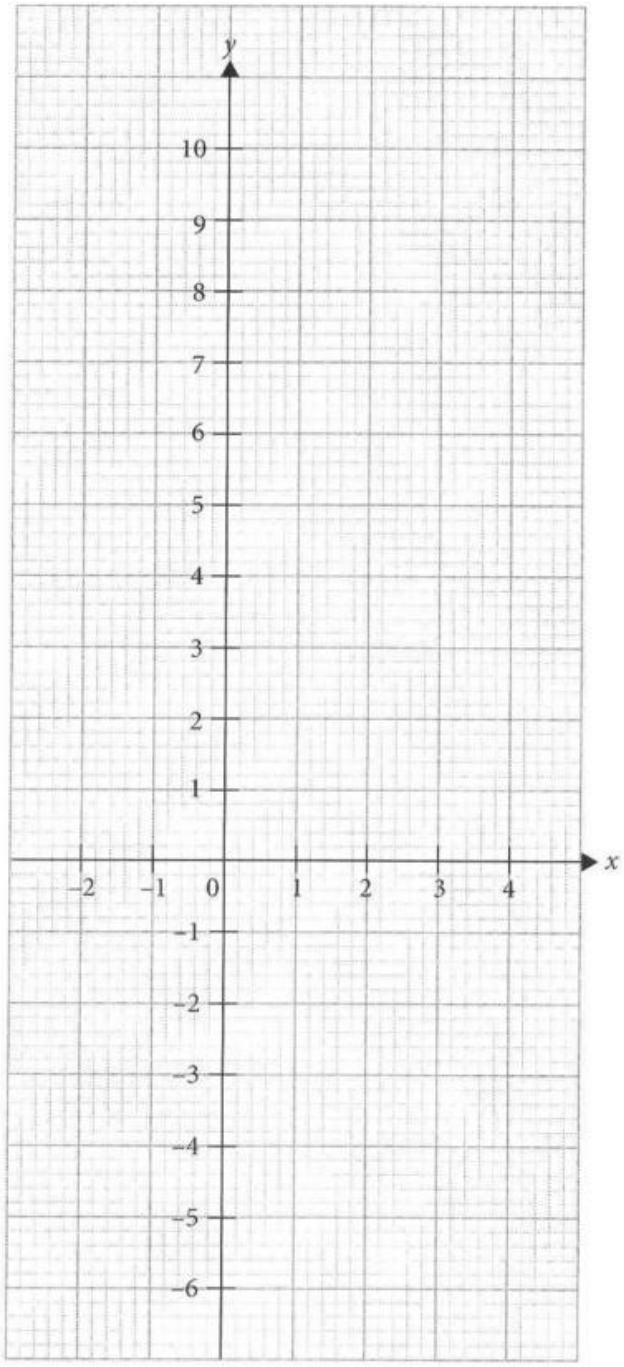
(c) Use your graph in part **(b)** to find

(i) the value of y when $x = 2.8$,

(ii) the values of x when $y = -3$,

(iii) the minimum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph.



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4. The variables x and y are connected by the equation $y = -x^2 + 2x + 1$.
Some corresponding values of x and y are given in the table below.

x	−1	−0.5	0	0.5	1	2	3
y	−2	p	1	1.75	2	1	−2

(a) Find the value of p.

(b) On the grid, draw the graph of $y = x^2 - 3x - 2$ for $-1 \leq x \leq 3$.

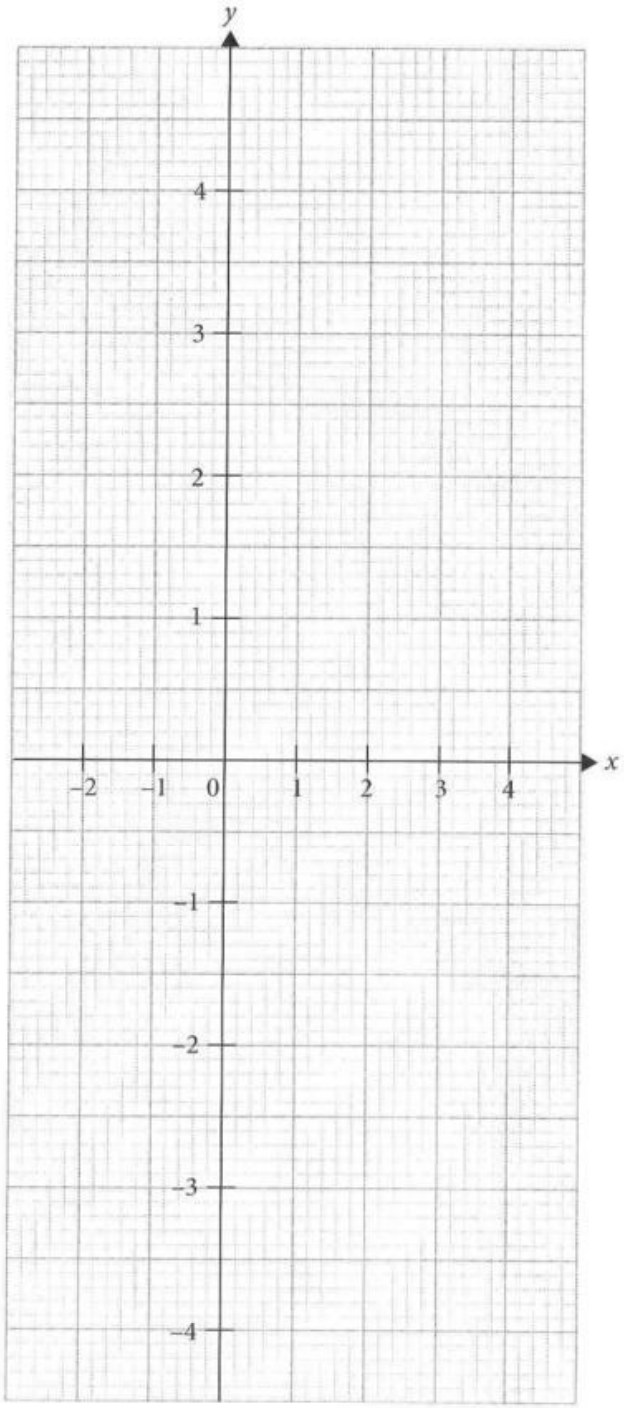
(c) Use your graph in part **(b)** to find

(i) the value of y when $x = -0.7$,

(ii) the values of x when $y = 0.5$,

(iii) the maximum value of y and the value of x at which this occurs.

(d) State the equation of the line of symmetry of the graph.

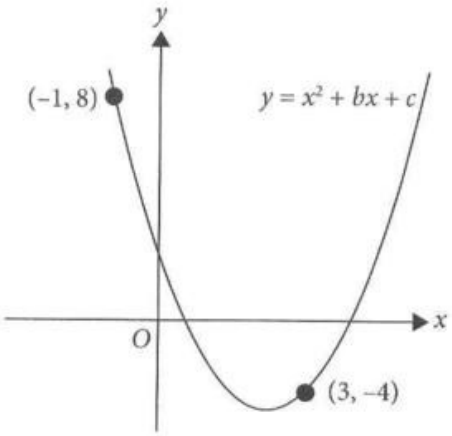


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Review Exercise 5

1. The curve $y = x^2 + bx + c$ passes through the points $(-1, 8)$ and $(3, -4)$.



(a) Find the values of b and c.

(b) Write down the coordinates of the point at which the curve intersects the y-axis.

---

2. Solve each of the following equations.

(a) $86x^2 = 0$

(b) $3 - \frac{1}{27}x^2 = 0$

(c) $12(x + 1)^2 = 17 - 4x$

(d) $x + 32 = \frac{1}{7}(x+7)(x-4)$

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3. It is given that $x = 9$ is a solution of the equation $7(x^2 + 3kx + 5) = (1 - k)x - 1$.
Find
(a) the value of k,
(b) the other solution of the equation.

---

4. **(i)** Solve the equation $8x^2 - 35x + 12 = 0$.
(ii) Hence, solve the equation $8(x + 4)^2 - 35(x + 4) + 12 = 0$.

---

5. The sum of the squares of three consecutive positive even numbers is 1208.
Find the three numbers.

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6. I think of a two-digit number.
The ones digit is 4 less than the tens digit.
The sum of the number and the product of its digits is 56.
Find the original number.

---

7. An open container is in the shape of a cuboid.
The height of the container is 22 cm less than its length.
The height of the container is also half of its width.
   (a) Given that the length of the container is x cm, write down expressions, in terms of x, for the height and the width of the container.
   (b) The external surface area of the open container is 1806 cm^2.
   Write down an equation to represent this information and show that it simplifies to
   $$4x^2 - 132x + 65 = 0.$$
   (c) Solve the equation $4x^2 - 132x + 65 = 0$.
   (d) Explain why one of the solutions in part (c) must be rejected as the length of the container.
   (e) Calculate the volume of the container.

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8. **(a)** Complete the table.

x	−4	−3	−2	−1	0	1	2
$y = 2 - x - 0.5x^2$							

(b) Using a scale of 2 cm to represent 1 unit on the x-axis and 1 cm to represent 1 unit on the y-axis, draw the graph of $y = 2 - x - 0.5x^2$ for $-4 \leq x \leq 2$.

(c) Use your graph in part **(b)** to find

(i) the coordinates of the turning point of the graph, stating whether it is a maximum point or a minimum point,

(ii) the equation of the line of symmetry of the graph.

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Worksheet 6A

Algebraic fractions

1. Simplify each of the following.

(a) $\frac{10ab}{100bc}$

(b) $\frac{7d^5}{21d^6}$

(c) $\frac{8h^2k}{2hk^2}$

(d) $\frac{6hj^3k^2}{9hj^4k}$

(e) $\frac{p^3(p+4q)}{3p(p+4q)^2}$

(f) $\frac{30p(2q+r)}{24p^2(2r+q)}$

(g) $\frac{15(x-y)^4}{35(x-y)^3}$

(h) $\frac{49x^3z(x^2-y)}{(7yz)^3(y-x^2)}$

---

2. Simplify each of the following.

(a) $\frac{a}{5a^2+a}$

(b) $\frac{b-b^2}{b}$

(c) $\frac{16h^2+2hk}{6h^2}$

(d) $\frac{3hj-3hk}{9hj-9hk}$

(e) $\frac{4p+4q}{(p+q)^2}$

(f) $\frac{7p-q}{8qr-56pr}$

(g) $\frac{22xy+33y^2}{99xy+66x^2}$

(h) $\frac{14y^3z-14xy^3}{7y(x-z)^2}$

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3. Simplify each of the following.

(a) $\frac{a^2-16b^2}{ab+4b^2}$

(b) $\frac{(6a-6b)^2}{6a^2-6b^2}$

(c) $\frac{5c+1}{5c^2+11c+2}$

(d) $\frac{d^2+3d-28}{d-4}$

(e) $\frac{h^2-9h+18}{h^2-6h}$

(f) $\frac{h^2+8hk+16k^2}{5h^2+18hk-8k^2}$

(g) $\frac{12m^2-10m-12}{24m^2-72m+54}$

(h) $\frac{15am-4n-5m+12an}{8n^2+6mn-5m^2}$

(i) $\frac{49p^2-28pq+4q^2}{8q^2-98p^2}$

(j) $\frac{9p^2-25q^2}{(3p+q)^2-16q^2}$

(k) $\frac{7x-xy+35-5y}{7x+5y-35-xy}$

(l) $\frac{wxy-wyz+xyz-yz^2}{wx^2-wxz-x^2z+xz^2}$

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4. **(i)** Factorise completely $50x^2 - 2$.

(ii) Simplify $\dfrac{50x^2-2}{10x^2-8x-2}$.

---

5. A student wrote the following.

$$\begin{aligned}\frac{3x^2-x}{9ax^3-6ax^2+ax} &= \frac{3\cancel{x^2}-\cancel{x}}{ax\left(9\cancel{x^2}-6\cancel{x}+1\right)}\\ &= \frac{3-1}{ax(9-6+1)}\\ &= \frac{\cancel{2}}{\cancel{4}ax}\\ &= 2ax\end{aligned}$$

Do you agree with the student? If you do not, identify the error(s) and write the correct working to simplify $\dfrac{3x^2-x}{9ax^3-6ax^2+ax}$.

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6. **(a)** Write down a possible algebraic fraction with two terms in the numerator and three terms in the denominator such that the fraction can be reduced to $\frac{4}{x-1}$.
Show your working clearly.

(b) Write down a possible algebraic fraction with two terms in the numerator and two terms in the denominator such that the fraction can be reduced to $\frac{4}{x-1}$.
Show your working clearly.

(c) Write down a possible algebraic fraction with four terms in the numerator and three terms in the denominator such that the fraction can be reduced to $\frac{x-2y}{x+3}$.
Show your working clearly.

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